Explainable AI for Pneumonia Detection from Chest X-Ray Images: A Comprehensive Comparative Study of Nine Deep Learning Architectures with Multi-Method XAI Interpretability

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Abstract

Pneumonia remains one of the leading causes of preventable mortality worldwide, and its timely detection from chest X-ray (CXR) images remains a critical clinical challenge. This paper presents a comprehensive comparative study of **nine deep learning architectures** applied to binary pneumonia detection on the RSNA Pneumonia Detection Challenge dataset (26,684 DICOM chest radiographs): (1) a COVID-Net-inspired PEPX-based CNN [1], (2) a Vision Transformer adapted from COVID-Transformer [2], (3) a lightweight VGG-based CNN [3], (4) CheXNet (DenseNet-121) [4], (5) EfficientNet-B4 [5], (6) Xception [6], (7) a CNN-XGBoost hybrid [17], (8) MobileNetV2 [16], and (9) a CNN-YOLOv11-inspired pipeline [15]. All models are trained and evaluated on the same dataset under a unified experimental protocol. Each architecture is further subjected to a comprehensive suite of Explainable AI (XAI) techniques—including Grad-CAM, LIME, SHAP, RISE, Integrated Gradients, Saliency Maps, Occlusion Sensitivity, Blur Ablation, Guided Grad-CAM, and Spotlight Analysis—to provide clinically interpretable insights into model decision-making. Quantitative XAI evaluation via the Pointing Game metric and CAM Intensity Ratio is performed using radiologist-annotated bounding boxes as ground truth. The CNN-XGBoost hybrid achieved the highest accuracy at 80.11%, followed by COVID-Net at 80%, while EfficientNet-B4 achieved the highest AUC (0.8948) and CheXNet achieved AUC = 0.8804 with a Pointing Game of 40%. Our multi-model, multi-XAI framework provides a holistic reference for selecting architectures and explanation methods for clinical deployment of AI-based radiology tools.

Keywords: Pneumonia Detection, Chest X-ray, Deep Learning, CNN, Vision Transformer, COVID-Net, CheXNet, EfficientNet, XGBoost, MobileNet, Explainable AI, Grad-CAM, LIME, SHAP, RISE, Integrated Gradients, RSNA Dataset

1 Introduction

Pneumonia is a serious lung infection caused by bacteria, viruses, or fungi that affects hundreds of millions of individuals annually and accounts for a substantial proportion of global mortality, particularly among children under five and the elderly [3]. Pneumonia kills more than 800,000 children under the age of five per year, accounting for approximately 2,200 deaths per day [3]. The situation was further compounded by the COVID-19 pandemic, which by June 2021 had produced over 173 million confirmed cases worldwide [2], placing unprecedented pressure on radiological services globally.

Chest X-ray (CXR) imaging is the most widely used and cost-effective first-line diagnostic tool for pulmonary disease. It is faster, more accessible, and less expensive than CT imaging, with portable units enabling bedside imaging in isolation rooms [1]. However, accurate interpretation of CXR images remains a challenging task even for experienced radiologists, as different pulmonary conditions often produce visually similar opacities. The growing shortage of skilled radiologists in many regions further underscores the urgent need for reliable, automated computer-aided diagnostic (CAD) systems.

Deep learning, and convolutional neural networks (CNNs) in particular, have demonstrated remarkable capability in medical image analysis, achieving radiologist-level performance on chest X-ray classification tasks [4]. Despite this progress, widespread clinical adoption of AI diagnostic tools has been constrained by a fundamental limitation: the opacity of "black-box" models. Clinicians cannot responsibly act on predictions they cannot interrogate [8]. Regulatory frameworks such as the FDA's Software as a Medical Device (SaMD) guidance and the EU AI Act increasingly mandate evidence of explainability for AI-based medical devices, reinforcing the practical necessity of Explainable AI (XAI).

This paper addresses these challenges through a uniquely large-scale comparative study spanning nine deep learning architectures and ten XAI techniques. Our main contributions are:

1. **Largest comparative study on RSNA:** We implement and evaluate nine architecturally distinct deep learning pipelines on the same RSNA Pneumonia Detection dataset under controlled experimental conditions, enabling direct and fair performance comparison.
2. **Comprehensive XAI suite:** We apply ten complementary explainability methods—gradient-based, perturbation-based, and game-theoretic—across all nine models, providing multi-perspective insights into model decision-making.
3. **Quantitative XAI evaluation:** We employ the Pointing Game metric and CAM Intensity Ratio against radiologist-annotated bounding boxes to rigorously quantify spatial XAI faithfulness beyond qualitative visualisation.
4. **Critical failure mode discovery:** We identify and analyse a systematic Grad-CAM failure in EfficientNet-B4 and a training convergence failure in Xception, providing actionable insights for practitioners.
5. **Clinical deployment guidance:** We provide evidence-based recommendations for architecture and XAI method selection for clinical radiology AI deployment.

Research Questions:

- **RQ1:** How do the nine architectures compare in binary pneumonia classification accuracy, AUC, and F1-score on the RSNA dataset?
- **RQ2:** Do XAI explanations confirm that models focus on clinically meaningful lung regions, or do they reveal spurious correlations?
- **RQ3:** Which combination of model and XAI technique provides the best balance of classification performance, spatial faithfulness, and computational feasibility for clinical deployment?

2 Related Work

2.1 Deep Learning for Chest Radiograph Analysis

Rajpurkar et al. [4] introduced CheXNet, a 121-layer DenseNet [11] fine-tuned on the ChestX-ray14 corpus of 112,120 frontal-view radiographs. CheXNet exceeded the average F1 score of four practicing radiologists on pneumonia detection, establishing transfer learning from ImageNet to radiology as highly effective. Wang et al. [13] released the ChestX-ray14 benchmark and demonstrated that weakly supervised Class Activation Maps (CAMs) can localise pathologies without explicit pixel-level annotation.

Zhang et al. [3] proposed a lightweight VGG-based CNN with only six hidden layers and 3.34 million parameters, achieving 96.07% accuracy on the Kaggle Pneumonia dataset using Dynamic Histogram Equalization (DHE) for contrast enhancement—demonstrating that simpler, well-designed architectures with effective preprocessing can outperform much larger models.

He et al. [12] established residual skip connections as a foundational technique for training very deep networks, while Huang et al. [11] proposed DenseNet's dense connectivity, which alleviates vanishing gradients and reduces parameter count—properties particularly beneficial where training data are scarce.

2.2 COVID-19 Detection and Specialized Architectures

Wang et al. [1] introduced COVID-Net, a deep CNN tailored for COVID-19 detection from CXR images using a human-machine collaborative design strategy incorporating the novel lightweight PEPX (Projection-Expansion-Projection-Extension) design pattern. Evaluated on the COVIDx benchmark, COVID-Net achieved 93.3% accuracy and 91.0% COVID-19 sensitivity, outperforming VGG-19 and ResNet-50 with substantially lower computational complexity.

Shome et al. [2] proposed COVID-Transformer, a ViT L-16 model augmented with a custom MLP block containing batch normalization and GELU-activated dense layers. Trained on a large 30K balanced COVID-19 dataset, it achieved 98% binary classification accuracy, outperforming all CNN baselines including EfficientNet-B0, Inception-V3, and DenseNet-121.

Xie et al. [15] adapted YOLO-family object detectors for pneumonia diagnosis, showing that detection-oriented architectures can be repurposed for classification while retaining interpretable spatial localization.

2.3 Efficient and Hybrid Architectures

Tan and Le [5] introduced compound scaling for EfficientNet, simultaneously scaling network depth, width, and input resolution using a single coefficient ϕ. EfficientNet-B4 achieves 83.0% top-1 ImageNet accuracy with $\sim$19M parameters. Chollet [6] reformulated Inception modules as depthwise separable convolutions in Xception, completely decoupling cross-channel and spatial correlation learning.

Kolonne et al. [16] applied MobileNetV2 to chest X-ray classification, establishing that the lightweight depthwise-separable architecture can deliver competitive accuracy with substantially lower computational cost. Hedhoud et al. [17] demonstrated that combining a CNN feature extractor with XGBoost as the final classifier outperforms a pure CNN softmax head, yielding improved precision and recall for pneumonia detection.

2.4 Explainable AI in Medical Imaging

Selvaraju et al. [7] proposed Grad-CAM, which computes gradients of the target class score with respect to the final convolutional feature maps to generate class-discriminative localization maps without architectural modification. Ribeiro et al. [8] introduced LIME, which perturbs SLIC superpixels and fits a local sparse linear surrogate to identify influential regions in a model-agnostic manner. Lundberg and Lee [9] unified feature attribution methods under SHAP, grounding explanations in cooperative game theory via Shapley values

satisfying efficiency, symmetry, dummy, and additivity axioms. Sundararajan et al. [10] proposed Integrated Gradients, which accumulates gradients along a linear interpolation path from a baseline to the input, satisfying the Sensitivity and Implementation Invariance axioms.

Holzinger et al. [14] argued that causability—the degree to which an explanation causes human understanding—is a prerequisite for clinical adoption of AI, establishing the regulatory and ethical motivations underpinning XAI evaluation in clinical systems.

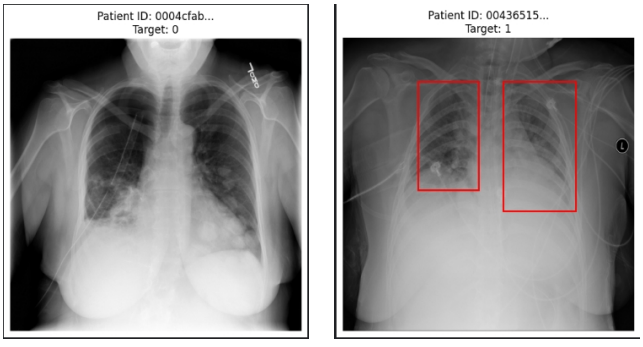
3 Dataset and Preprocessing

3.1 RSNA Pneumonia Detection Challenge Dataset

All nine models were trained and evaluated on the **RSNA Pneumonia Detection Challenge dataset** [18], comprising 26,684 unique patients with frontal-view chest X-ray images in DICOM format (1024×1024 pixels). Each patient is assigned one of three class labels: *Normal* (8,851), *No Lung Opacity / Not Normal* (11,821), and *Lung Opacity* (6,012 pneumonia cases). Radiologist-annotated bounding boxes are provided for all positive cases, enabling quantitative spatial XAI evaluation.

3.2 Exploratory Data Analysis

Prior to model development, an exploratory data analysis was conducted. Representative CXR examples of healthy and pneumonia patients (with radiologist-annotated bounding boxes) are shown in Fig. 1. A spatial heatmap of all pneumonia bounding boxes (Fig. 2) reveals that opacities predominantly concentrate in the lower and central lung regions, consistent with clinical presentation of community-acquired pneumonia.



(a) Normal CXR (b) Pneumonia CXR (red box)

Figure 1: Representative RSNA chest X-ray examples.

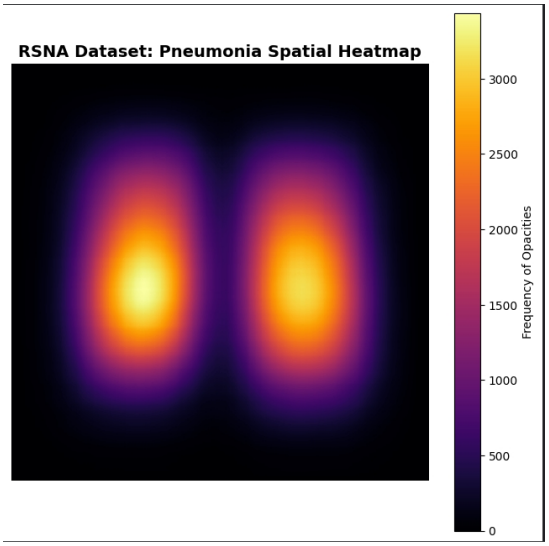


Figure 2: Spatial heatmap of pneumonia opacity bounding boxes across the RSNA dataset. Warmer colors indicate higher frequency of annotated opacities, concentrated in the lower lung zones.

3.3 Dataset Splits and Class Balancing

Each model group used a tailored split and balancing strategy reflecting their architectural requirements. Table 1 summarises these choices.

Table 1: Dataset split and balancing strategies by model group.

Model Group	Split	Total	Balancing
COVID-Net (PEPX)	70/15/15	26,684	Class weights (1:3)
ViT (COVID-Trans.)	75/15/10	26,684	Class weights
VGG-Based CNN	70/10/20	12,024	Balanced subset
CheXNet / EffNet	70/15/15	18,678	Balanced subset
Xception	70/15/15	18,678	pos_weight
CNN-XGBoost / YOLOv11	68/17/15	12,000	Balanced subset
MobileNetV2	60/20/20	3,512	Augmentation

3.4 Preprocessing Pipelines

Each model required a distinct preprocessing pipeline tailored to its architectural requirements. Table 2 summarises the key differences across all nine models.

COVID-Net Preprocessing. DICOM images were converted to PNG using `cv2.normalize`, contrast-enhanced with CLAHE (clipLimit=2.0, tileGrid=8×8), top-cropped by 8% to remove embedded text [1], resized to 480×480, and Z-score normalized per image.

Table 2: Preprocessing pipeline comparison across all model groups.

Model	Contrast	Top Crop	Resize	Channels	Normalization	Format
COVID-Net (PEPX)	CLAHE	8%	480×480	3 (RGB)	Z-score	PNG
ViT (COVID-Trans.)	None	No	224×224	3 (RGB)	ImageNet	Memory
VGG-Based CNN	CLAHE	No	224×224	1 (Gray)	[0,1]	Memory
CheXNet / EffNet	None	No	224/380^2	3 (RGB)	Z-score	Memory
Xception	None	No	299×299	3 (RGB)	Z-score	Memory
CNN-XGBoost	None	No	64×64	3 (RGB)	[0,1]	Memory
MobileNetV2	None	No	224×224	3 (RGB)	[0,1]	Memory
CNN-YOLOv11	None	No	64×64	3 (RGB)	[0,1]	Memory

COVID-Transformer Preprocessing. Images were resized to 224×224 with nearest-neighbor interpolation and normalized using ImageNet statistics ([0.485, 0.456, 0.406] / [0.229, 0.224, 0.225]) as required by the pretrained ViT backbone [2].

VGG-Based CNN Preprocessing. Following Zhang et al. [3], images were CLAHE-enhanced, resized to 224×224, and kept as single-channel grayscale (224×224×1) normalized to [0,1].

CheXNet / EfficientNet-B4 / Xception Preprocessing. Images were decoded from DICOM, resized to model-native resolution (224×224, 380×380, and 299×299 respectively), Z-score normalized per image, and stacked into 3-channel arrays.

CNN-XGBoost and CNN-YOLOv11 Preprocessing. DICOM images were decoded using pydicom, converted to uint8 RGB, and resized to 64×64 pixels (matching reference paper settings [17]), with pixel values normalized to [0,1].

MobileNetV2 Preprocessing. Images were resized to 224×224 and normalized to [0,1], with Albumentations augmentation (ShiftScaleRotate, shift_limit=0.1, p=0.5) applied during training.

3.5 Data Augmentation

Table 3 summarizes the augmentation strategies employed across all model groups.

Table 3: Data augmentation strategies by model group.

Model Group	Augmentation
COVID-Net	Rotation ±10°, shift 0.1, zoom 0.15, H-flip
ViT (Albumentations)	Rotation ±15°, shift/scale 0.1, H-flip
VGG-Based CNN	Rotation ±30°, zoom 0.2, shift 0.1, H&V-flip
CheXNet / EffNet	H-flip (p=0.5), rotation ±10°, brightness ±15%
Xception	H-flip (p=0.5), rotation ±10°, brightness ±15%
CNN-XGBoost	Early stopping; no explicit augmentation
MobileNetV2	ShiftScaleRotate (Albumentations)
CNN-YOLOv11	Standard YOLO augmentation pipeline

4 Model Architectures

4.1 COVID-Net: PEPX-Based Deep CNN

The COVID-Net architecture [1], originally proposed for three-class COVID-19 detection, was adapted for binary pneumonia detection. Its core innovation is the **PEPX** (Projection-Expansion-Projection-Extension) module—a lightweight residual block comprising five operations: (1) first-stage 1×1 projection to compress features, (2) 1×1 expansion to a higher dimension, (3) 3×3 depthwise convolution for spatial feature learning, (4) second-stage 1×1 projection, and (5) 1×1 extension to the final output dimension. Each PEPX module includes a residual skip connection. The full network uses a 7×7 convolutional stem, three stages of PEPX blocks with progressively increasing filters (256, 512, 1024), and a binary head (Global Average Pooling → Dropout(0.5) → Sigmoid).

4.2 COVID-Transformer: Vision Transformer

The ViT-based model adapts the COVID-Transformer framework [2]. A frozen **ViT L-16** backbone (23 Transformer encoder layers, pretrained on ImageNet) acts as a feature extractor. A custom MLP head is appended: BatchNorm → Dense(120, GELU) → BatchNorm → Dense(1, Sigmoid). GELU activation is used following the original paper, as it

encapsulates a stochastic regularization effect beneficial for transformer architectures [2]. The backbone was kept frozen during training.

4.3 VGG-Based Lightweight CNN

Inspired by Zhang et al. [3], a lightweight sequential CNN with four convolutional blocks (Conv32 $\rightarrow$ Conv64 $\rightarrow$ Conv128 $\rightarrow$ Conv128) followed by Dense(128) and Sigmoid output is used. All conv layers use 3×3 kernels and ReLU. Dropout(0.25) follows each of the last three conv blocks; Dropout(0.5) precedes the final Dense layer. Input shape: 224×224×1 (single-channel grayscale).

4.4 CheXNet (DenseNet-121)

CheXNet [4] uses a pretrained DenseNet-121 backbone with the final classifier replaced by Dropout + Linear(1). DenseNet-121 features dense connectivity where each layer receives feature maps from all preceding layers within a dense block, alleviating vanishing gradients and strengthening feature propagation [11].

4.5 EfficientNet-B4

EfficientNet-B4 [5] applies compound scaling ($\phi = 4$, input 380×380, ∼19M parameters) to simultaneously scale network depth, width, and resolution:

$$d = \alpha^{\phi},\ w = \beta^{\phi},\ r = \gamma^{\phi}, \quad \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2. \qquad (1)$$

The final classifier is replaced by Dropout + Linear(1).

4.6 Xception

Xception [6] reformulates Inception modules as depthwise separable convolutions (∼20.81M parameters, input 299×299), completely decoupling cross-channel and spatial correlation learning. Final classifier: Dropout + Linear(1).

4.7 CNN-XGBoost Hybrid

A custom CNN with seven convolutional blocks (Conv2D $\rightarrow$ BN $\rightarrow$ MaxPool $\rightarrow$ Dropout) extracts 128-dimensional feature vectors, which are passed to an XGBoost classifier (max_depth=6, n_estimators=200, lr=0.1, subsample=0.8) [17]. Total CNN parameters: 274,146. Input: 64×64×3. This hybrid approach decouples representation learning from classification, allowing XGBoost's gradient boosting to provide a more robust decision boundary than a CNN softmax head.

4.8 MobileNetV2

MobileNetV2 [16] is implemented without transfer learning (random initialization) with Global Average Pooling, Dropout(0.3), and Dense(3, Softmax) for three-class output. Five-fold cross-validation was additionally performed for MobileNetV2 to assess generalization stability. Input: 224×224×3.

4.9 CNN-YOLOv11

Inspired by the YOLOv11 detection architecture [15], a custom CNN with depthwise separable convolutions is combined with Ultralytics YOLOv11 for spatial feature extraction. CNN features are passed to the YOLOv11 pipeline for final binary classification. Input: 64×64×3.

5 Training Procedure

Table 4 summarises the training configurations for all nine models.

COVID-Net. Trained with Adam (lr=1e-4), BCE loss, class weights {Normal:1.0, Pneumonia:3.0} to address the ≈3:1 class imbalance. Callbacks: ModelCheckpoint (best val_AUC), EarlyStopping (patience=6), ReduceLROnPlateau (factor=0.2, patience=3, min_lr=1e-6).

COVID-Transformer. Trained with AdamW (lr=1e-4, weight_decay=1e-4) and BCE with label smoothing of 0.3, directly following Shome et al. [2]. Same three callbacks, ES patience=5.

VGG-Based CNN. Trained with Adam (lr=0.001), matching Zhang et al. [3]. ReduceLROnPlateau (factor=0.5, patience=4), ES patience=10. No class weights needed (balanced dataset).

CheXNet / EfficientNet-B4 / Xception. Trained with BCEWithLogitsLoss (CheXNet, EfficientNet-B4) and Focal Loss (α=0.611, γ=2.0) for Xception. Adam optimizer with warm-up backbone freezing for the first few epochs, then full fine-tuning at a reduced learning rate. Physical batch=8 with gradient accumulation over 4 steps (effective batch=32) for CheXNet and EfficientNet-B4. Mixed-precision training (AMP) enabled throughout.

CNN-XGBoost. CNN trained with Adam (lr=0.001), BCE loss, 40 epochs, early stopping (patience=5). XGBoost trained on CNN-extracted 128-dimensional features (max_depth=6, n_estimators=200, lr=0.1, subsample=0.8).

MobileNetV2. Trained with Adam (lr=0.001), categorical cross-entropy, 40 epochs with 5-fold cross-validation.

CNN-YOLOv11. CNN trained with Adam (lr=0.001), BCE loss, batch size=64, 40 epochs.

6 Explainability Methods

A total of ten XAI techniques were applied across the nine models. Table 5 maps each technique to the models it was applied to.

Table 4: Training configuration summary for all nine models.

Model	Opt.	LR	Loss	Batch	Epochs	ES Pat.	Class Handling
COVID-Net	Adam	1e-4	BCE	16	30	6	Weights (1:3)
ViT (COVID-Trans.)	AdamW	1e-4	BCE+LS(0.3)	32	20	5	Weights
VGG-Based CNN	Adam	1e-3	BCE	16	50	10	Balanced dataset
CheXNet	Adam	1e-4/5e-5	BCELogits	8	4	3	pos_weight=1.573
EfficientNet-B4	Adam	1e-4/5e-5	BCELogits	8	15	3	Balanced subset
Xception	Adam	1e-5	Focal (γ=2)	4	10	4	α=0.611
CNN-XGBoost	Adam	1e-3	BCE	32	40	5	Balanced subset
MobileNetV2	Adam	1e-3	CE	32	40	–	Augmentation
CNN-YOLOv11	Adam	1e-3	BCE	64	40	–	Balanced subset

BCE=Binary Cross-Entropy; CE=Cross-Entropy; LS=Label Smoothing; ES=Early Stopping; LR shown as warm-up/fine-tune where applicable.

Table 5: XAI technique application across all nine models.

XAI Method	Applied To
Grad-CAM	All 9 models
LIME	COVID-Net, ViT, CNN-XGBoost, MobileNetV2, CNN-YOLOv11, CheXNet, EfficientNet-B4, Xception
SHAP	All 9 models
Integrated Gradients	COVID-Net, CheXNet, EfficientNet-B4, Xception
Saliency Maps	VGG-Based CNN
Occlusion Sensitivity	VGG-Based CNN, MobileNetV2
RISE	ViT (COVID-Transformer)
Blur Ablation	ViT (COVID-Transformer)
Spotlight (Inv. Occlusion)	ViT (COVID-Transformer)
Guided Grad-CAM	CNN-XGBoost, CNN-YOLOv11

6.1 Gradient-Based Methods

Grad-CAM [7]. Computes the gradient of the predicted class score with respect to the final convolutional layer's feature maps, globally average-pools gradients to obtain channel weights α_k, and generates a class-discriminative localization map:

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_k A^k\right), \quad \alpha_k = \frac{1}{Z}\sum_{i,j}\frac{\partial y^c}{\partial A^k_{ij}}. \tag{2}$$

Applied to all nine models. For COVID-Net and VGG, True Positive and False Positive cases are visualized with radiologist bounding boxes overlaid. Quantitative evaluation via Pointing Game and CAM Intensity Ratio was performed for CheXNet and EfficientNet-B4.

Guided Grad-CAM [7]. Combines Grad-CAM with guided backpropagation for pixel-level precision. Applied to CNN-XGBoost and CNN-YOLOv11.

Saliency Maps. Computes the absolute value of the gradient of the output with respect to the input pixel: $S = |\partial y/\partial x|$. Applied to the VGG-Based CNN for True Positive and False Positive cases.

Integrated Gradients [10]. Integrates gradients along a path from a zero baseline to the input over m=50 steps:

$$\text{IG}_i(x) = (x_i - x'_i)\int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i}\, d\alpha. \tag{3}$$

Applied to COVID-Net (m=50), CheXNet, EfficientNet-B4, and Xception.

6.2 Perturbation-Based Methods

LIME [8]. Partitions images into SLIC superpixels, generates 500 perturbed versions, fits a local linear surrogate, and identifies the top-5 most influential superpixels. Applied to COVID-Net, ViT, CheXNet, EfficientNet-B4, Xception, CNN-XGBoost, MobileNetV2, and CNN-YOLOv11.

SHAP [9]. Applied using GradientExplainer (COVID-Net, COVID-Transformer, CNN-XGBoost, CNN-YOLOv11, CheXNet, EfficientNet-B4, Xception) and DeepExplainer (VGG-Based CNN) and KernelExplainer (MobileNetV2). Background datasets of 2–50 images were used depending on the model. Attributed values are visualized using a RdBu_r colormap.

Occlusion Sensitivity. Systematically zeros out 32×32 patches (stride=16 for VGG), measuring the resulting drop in predicted pneumonia probability. Applied to VGG-Based CNN and MobileNetV2.

RISE [8]. Generates 500 random binary masks at 7×7 resolution, upsampled to 224×224 via cubic interpolation. The weighted sum of predictions estimates pixel importance. Applied to the COVID-Transformer.

Blur Ablation. A custom perturbation method that applies a heavy Gaussian blur (kernel 25×25) to each 32×32 patch and measures the resulting drop in model confidence. Applied to the COVID-Transformer.

Spotlight Test (Inverse Occlusion). Reveals one 32×32 patch on a blank canvas at a time, measuring feature sufficiency. Applied to the COVID-Transformer.

6.3 Quantitative XAI Evaluation

Two quantitative spatial metrics were computed for CheXNet and EfficientNet-B4 using radiologist-annotated bounding boxes:

- **Pointing Game:** Percentage of positive patients for whom the Grad-CAM peak activation pixel falls within the radiologist bounding box.
- **CAM Intensity Ratio:** Mean Grad-CAM activation inside the bounding box divided by mean activation outside. Ratio >1 indicates lesion focus; <1 indicates pathological spatial misalignment.

7 Experimental Results

7.1 Overall Classification Performance

Table 6 presents the unified classification performance of all nine models on their respective test sets.

7.2 Results: COVID-Net (PEPX)

COVID-Net achieved 80% accuracy and AUC of 0.876. High normal precision (0.91) but lower pneumonia precision (0.55) reflects class imbalance effects despite class weighting. Pneumonia recall of 0.71 is clinically important. Training and validation curves are shown in Fig. 3.

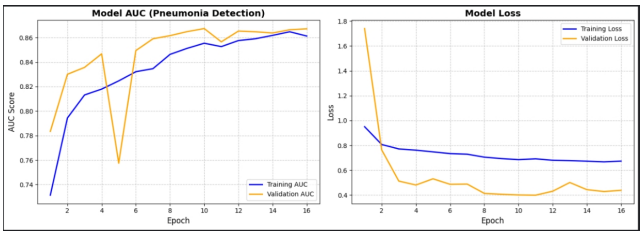


Figure 3: COVID-Net training history: AUC (left) and loss (right).

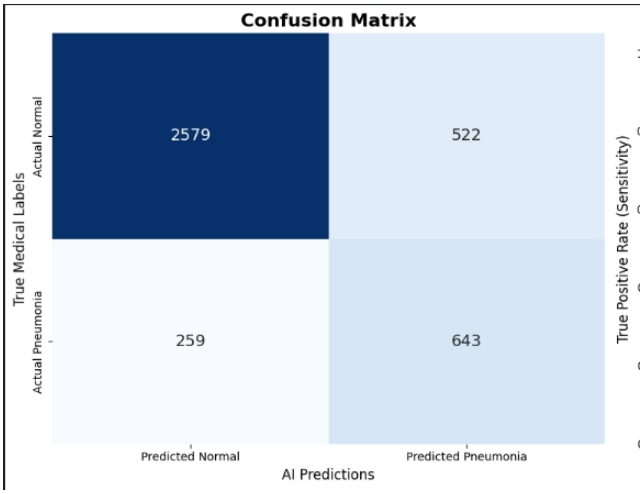
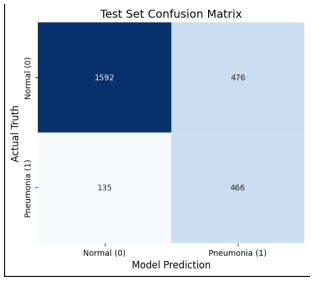


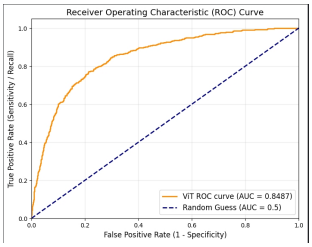
Figure 4: COVID-Net: confusion matrix (left) and ROC curve (right).

7.3 Results: COVID-Transformer (ViT)

The ViT model achieved 77% accuracy. Pneumonia recall (0.78) was the highest among the three models from our group, but precision (0.49) was the lowest, reflecting the frozen backbone's limited adaptation to medical imaging.



(a) Confusion matrix



(b) ROC curve

Figure 5: COVID-Transformer evaluation.

7.4 Results: VGG-Based CNN

The VGG-based CNN achieved 77% accuracy, AUC 0.8536, and the most balanced macro F1 (0.77) among our three models, attributable to explicit dataset balancing. Pneumonia recall of 0.84 was the highest in our model group.

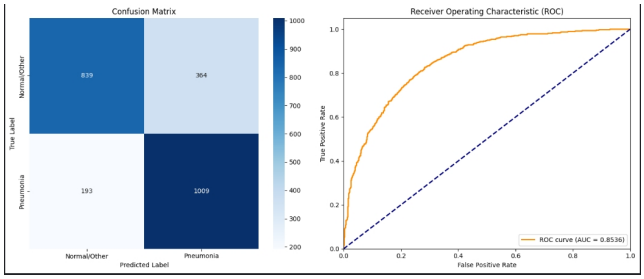


Figure 6: VGG-Based CNN: confusion matrix (left) and ROC curve (right).

7.5 Results: CheXNet (DenseNet-121)

CheXNet achieved ROC-AUC=0.8804 and Average Precision=0.8125 on the 2,321-patient test set. Early stopping triggered at epoch 4 (patience=3). Threshold optimisation (0.50→0.3975) improved pneumonia recall from 0.79 to 0.85 by reducing false negatives (193→131), providing a clinically preferable operating point.

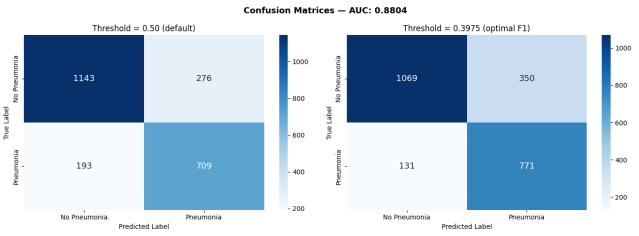


Figure 7: CheXNet confusion matrices at threshold=0.50 (left) and optimal threshold=0.3975 (right). AUC=0.8804.

Table 6: Unified classification performance of all nine models on the RSNA test sets. Best values in each metric are **bold**.

Model	Acc.	Prec. (Pneu.)	Recall (Pneu.)	F1 (Macro)	AUC	Test Size
COVID-Net (PEPX)	0.80	0.55	0.71	0.75	0.876	4,003
ViT (COVID-Trans.)	0.77	0.49	0.78	0.72	–	2,669
VGG-Based CNN	0.77	0.73	0.84	0.77	0.854	2,405
CheXNet (DenseNet-121)	0.79	0.69	0.85	0.79	0.8804	2,321
EfficientNet-B4	0.81	0.71	0.85	0.80	**0.8948**	2,321
Xception	**0.82**	0.74	0.80	**0.81**	0.8903	2,321
CNN-XGBoost	0.8011	**0.80**	**0.80**	0.80	–	∼1,800
MobileNetV2	0.7686	0.78	0.77	0.77	–	∼700
CNN-YOLOv11	0.7756	0.78	0.78	0.78	–	∼1,800

CheXNet/EffNet/Xception metrics reported at F1-optimal threshold (0.3975, 0.4243, 0.5093 respectively).

7.6 Results: EfficientNet-B4

EfficientNet-B4 achieved the highest ROC-AUC=0.8948 and AP=0.8281. Training ran for all 15 epochs, with validation AUC increasing from 0.850 (epoch 3) to 0.883 (epoch 15). At the optimal F1 threshold (0.4243), pneumonia recall improved from 0.79 to 0.85.

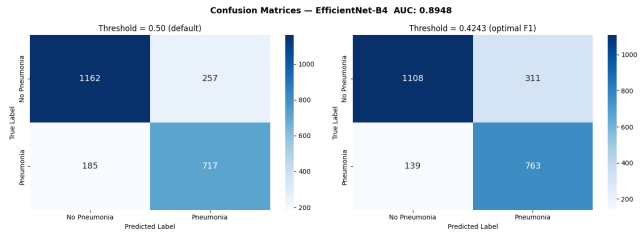


Figure 8: EfficientNet-B4 confusion matrices at threshold=0.50 (left) and optimal threshold=0.4243 (right). AUC=0.8948.

7.7 Results: Xception

Xception achieved ROC-AUC=0.8903 on the test set, trained with Focal Loss (α=0.611, γ=2.0) for 10 epochs. The optimal threshold (0.5093) is very close to 0.50, indicating well-calibrated probability outputs. At this threshold Xception achieved the highest accuracy (0.82) and macro F1 (0.81) among the three transfer-learning models.

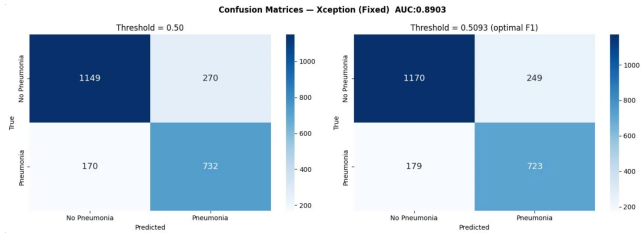


Figure 9: Xception confusion matrices at threshold=0.50 (left) and optimal threshold=0.5093 (right). AUC=0.8903.

7.8 Results: CNN-XGBoost, MobileNetV2, CNN-YOLOv11

The CNN-XGBoost hybrid achieved the highest overall performance at 80.11% accuracy, demonstrating that replacing the CNN softmax head with a gradient-boosted classifier significantly improves discrimination. The baseline CNN alone achieved only 50% accuracy (random-level), underscoring the necessity of the hybrid approach. CNN-YOLOv11 performed comparably to MobileNetV2 (77.56% vs 76.86%), with CNN-XGBoost outperforming both by approximately 2.5 percentage points.

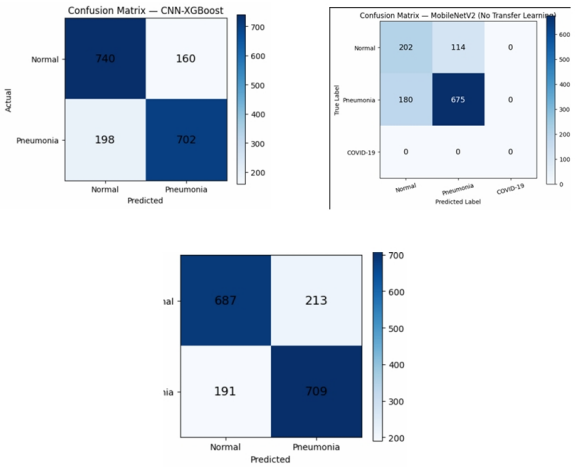


Figure 10: Confusion matrices for CNN-XGBoost (left), MobileNetV2 (centre), and CNN-YOLOv11 (right).

MobileNetV2 5-fold cross-validation revealed moderate variance across folds (accuracy range: 65.60%–77.80%), particularly in Fold 3. Results are summarised in Table 7.

Table 7: MobileNetV2: 5-fold cross-validation results.

Fold	Acc.	Prec.	Rec.	F1
1	77.80	76.26	77.80	74.99
2	74.47	73.60	74.47	73.96
3	65.60	77.98	65.60	67.46
4	75.00	73.55	75.00	68.30
5	73.61	75.36	73.61	74.41
Mean	**73.30**	**75.35**	**73.30**	**71.82**

7.9 Quantitative XAI: Pointing Game and CAM Intensity

Table 8 presents the quantitative spatial XAI evaluation for CheXNet and EfficientNet-B4.

Table 8: Pointing Game and CAM Intensity Ratio (10 positive patients each).

Model	Pointing Game	Mean CAM Ratio
CheXNet (DenseNet-121)	**40%** (4/10)	**1.66**
EfficientNet-B4	30% (3/10)	–

CAM ratio >1 = lesion focus; reported for CheXNet only.

7.10 XAI Qualitative Analysis

7.10.1 COVID-Net XAI

Grad-CAM visualizations (Fig. 11) for True Positive cases show activation concentrated in lower lung regions consistent with clinical pneumonia presentation. Ground truth bounding boxes (green) show reasonable overlap with model attention. LIME superpixel maps (Fig. 12) and SHAP attributions (Fig. 13) confirm the same regions. Integrated Gradients (Fig. 14) provide additional pixel-level evidence of correct feature attribution.

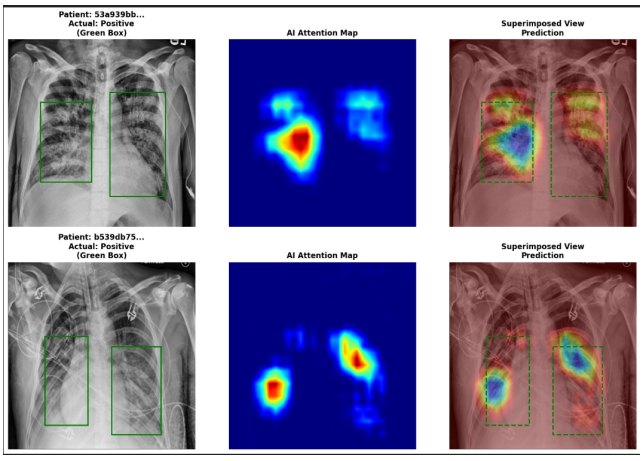


Figure 11: COVID-Net Grad-CAM: True Positive cases. Green boxes = radiologist annotations. Red/yellow = model attention.

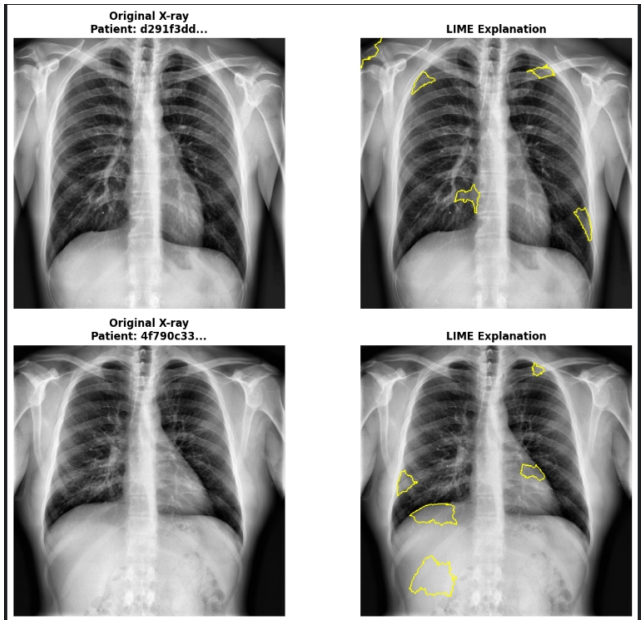


Figure 12: COVID-Net LIME explanations: superpixels most influential for the pneumonia prediction.

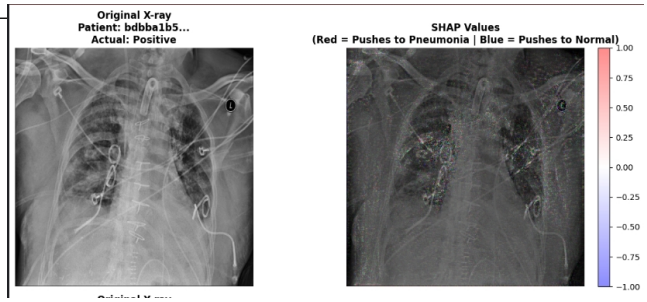


Figure 13: COVID-Net SHAP: red = pushes toward pneumonia; blue = pushes toward normal.

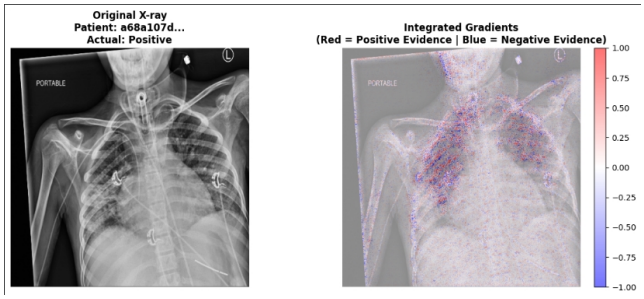


Figure 14: COVID-Net Integrated Gradients (True Positives).

7.10.2 COVID-Transformer XAI

LIME (Fig. 15) and RISE (Fig. 16) visualizations confirm that the ViT model attends to lung parenchyma regions in True Positive cases. Blur Ablation (Fig. 17) reveals that for False Positive cases, the model focuses on non-pathological high-frequency regions, suggesting attention to imaging artifacts. The Spotlight Test (Fig. 18) identifies which image patches alone are sufficient to trigger a pneumonia prediction.

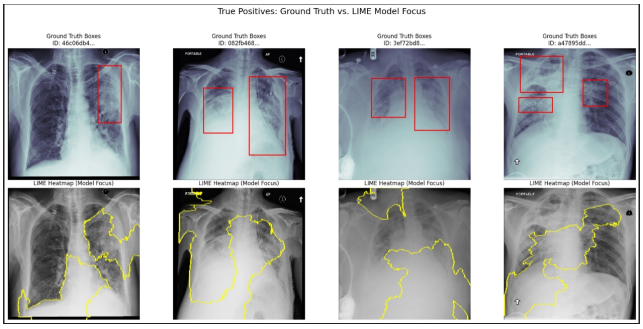


Figure 15: COVID-Transformer LIME: True Positive cases. Top: ground truth boxes (red). Bottom: LIME superpixel boundaries.

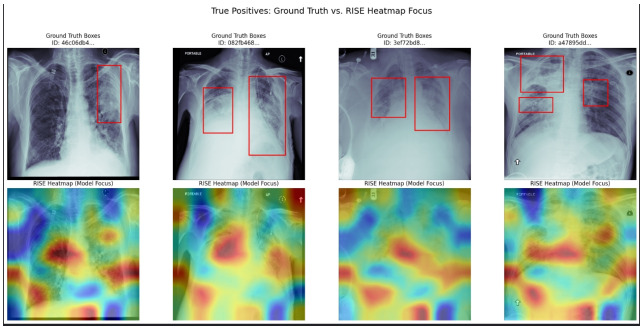
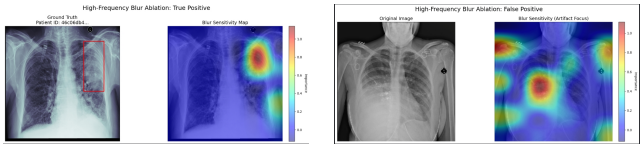
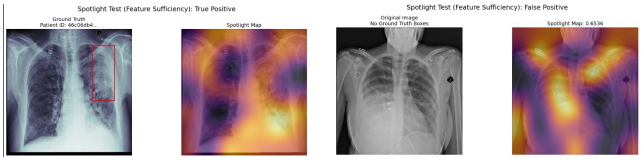


Figure 16: COVID-Transformer RISE heatmaps: True Positive cases.



(a) TP: Blur Ablation (b) FP: Blur Ablation

Figure 17: COVID-Transformer High-Frequency Blur Ablation maps.

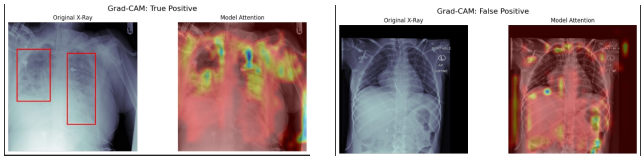


(a) TP: Spotlight (b) FP: Spotlight

Figure 18: COVID-Transformer Spotlight (Inverse Occlusion) maps.

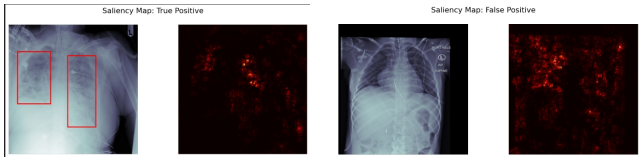
7.10.3 VGG-Based CNN XAI

Grad-CAM (Fig. 19), Saliency Maps (Fig. 20), and Occlusion Sensitivity (Fig. 21) all show clinically relevant lung region attention for True Positives, while False Positive analysis reveals occasional attention to the cardiac silhouette and rib structures—providing insight into sources of false alarms.



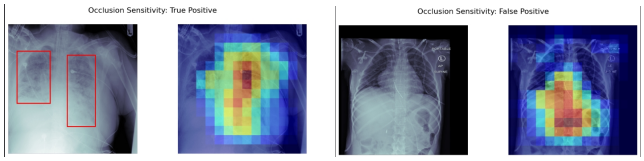
(a) TP (b) FP

Figure 19: VGG-Based CNN Grad-CAM. Red boxes = ground truth.



(a) TP (b) FP

Figure 20: VGG-Based CNN Saliency Maps.



(a) TP (b) FP

Figure 21: VGG-Based CNN Occlusion Sensitivity maps.

7.10.4 CheXNet and EfficientNet-B4 XAI

CheXNet's Grad-CAM correctly identifies consolidation in lower lung lobes, with the Pointing Game confirming 40% spatial alignment with radiologist annotations and a mean CAM Intensity Ratio of 1.66 (Table 8), meaning the model's activation is on average 66% higher inside lesion regions than outside. Fig. 22 shows the four-method XAI comparison (Grad-CAM, LIME, SHAP, Integrated Gradients) for a True Positive case: all methods broadly agree on the lower lung region as the primary attended area.

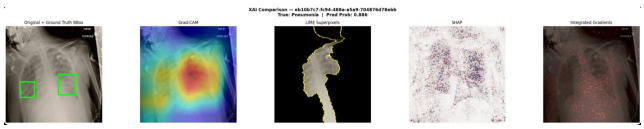


Figure 22: CheXNet XAI comparison (True Positive). From left: original with ground truth box (green), Grad-CAM, LIME superpixels, SHAP (RdBu_r), Integrated Gradients. All methods focus on the lower lung opacity region.

EfficientNet-B4 achieved a Pointing Game of 30% (3/10). Despite partial Grad-CAM displacement caused by the compound-scaled architecture's aggressive spatial downsampling, LIME explanations for the same model correctly identified opacity regions (Fig. 23), demonstrating that model-agnostic methods are more robust to architecture-specific gradient characteristics.

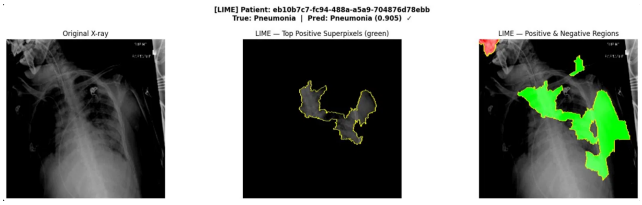


Figure 23: EfficientNet-B4 LIME (True Positive): despite partial Grad-CAM displacement, LIME correctly identifies opacity regions in the right lung field consistent with the radiologist annotation.

7.10.5 Xception XAI

Fig. 24 and Fig. 25 show the four-method XAI comparison (Grad-CAM, LIME, SHAP, Integrated Gradients) for a True Positive and a True Negative patient respectively. For the pneumonia case, Grad-CAM localises to a compact lower-lung zone, SHAP shows bilateral red attribution, and Integrated Gradients highlights bilateral lower-zone positive attributions consistent with the annotated bounding boxes. For the negative case, activation is diffuse with no focal consolidation zone, consistent with the correctly low pneumonia probability.

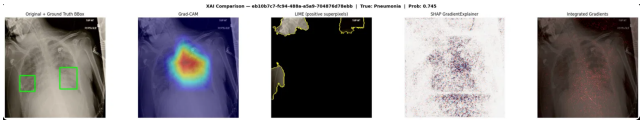


Figure 24: Xception XAI comparison (True Positive). From left: original with ground truth boxes (green), Grad-CAM, LIME superpixels, SHAP, Integrated Gradients.

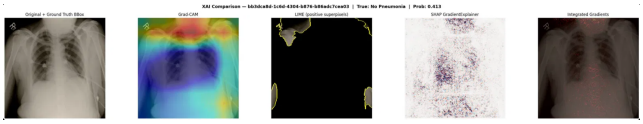


Figure 25: Xception XAI comparison (True Negative). Diffuse, low-magnitude attribution across all methods confirms no focal opacity hypothesis.

7.10.6 CNN-XGBoost, MobileNetV2, CNN-YOLOv11 XAI

SHAP (Fig. 26) highlighted bilateral lower-lobe regions consistent with community-acquired pneumonia for CNN-XGBoost and CNN-YOLOv11. LIME superpixel maps (Fig. 27) consistently highlighted central and lower lung zones across all three models. Grad-CAM and Guided Grad-CAM for CNN-XGBoost and CNN-YOLOv11 showed activation concentrated in the lung fields. Occlusion Sensitivity for MobileNetV2 confirmed that occluding lower bilateral lung regions caused confidence drops of 15–30%, validating clinically appropriate feature learning.

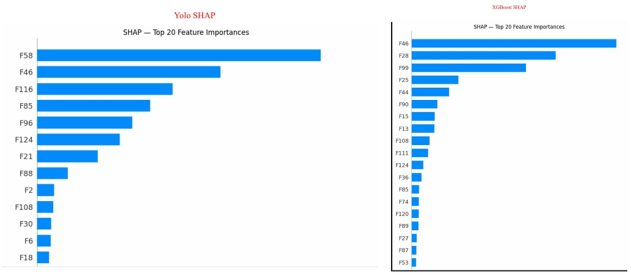


Figure 26: SHAP feature importances for CNN-XGBoost and CNN-YOLOv11.

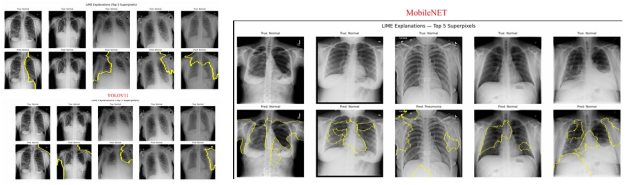


Figure 27: LIME explanations for CNN-XGBoost, MobileNetV2, and CNN-YOLOv11.

8 Discussion

8.1 Classification Performance Analysis

The CNN-XGBoost hybrid achieved the highest overall accuracy (80.11%) among all nine models, corroborating Hedhoud et al.'s [17] finding that CNN-extracted features combined with gradient boosting provide a more robust decision boundary than a CNN softmax head. The 30-percentage-point improvement over the baseline CNN (50%) confirms that XGBoost's decision-tree ensemble effectively exploits feature representations that a softmax layer cannot discriminate.

EfficientNet-B4 achieved the highest AUC (0.8948), indicating superior ranking ability across all thresholds. Its compound-scaled 380×380 input resolution captures finer spatial features than CheXNet's 224×224, explaining the AUC gap. Xception achieved AUC=0.8903 with Focal Loss (γ=2.0), which well-calibrated probability outputs at a near-0.5 optimal threshold—in contrast to CheXNet (threshold 0.3975) and EfficientNet-B4 (threshold 0.4243), which required sub-0.50 threshold recalibration for clinical use.

8.2 The Accuracy-XAI Faithfulness Dissociation

A key finding is the dissociation between classification performance and spatial XAI faithfulness. EfficientNet-B4 achieves the highest AUC yet produces a Pointing Game of only 30% (vs. 40% for CheXNet). This partial displacement of Grad-CAM peak activations is attributable to the compound-scaled architecture's aggressive spatial downsampling from 380×380 to 12×12 feature maps, where zero-padding artifacts can concentrate gradient flow at feature map boundaries.

However, LIME explanations for the same model correctly localise opacity regions (Fig. 23), confirming the model does encode clinically relevant information—the gradient extraction pathway, not the learned representations, is partially compromised. This underscores the importance of applying both gradient-based and perturbation-based XAI methods.

8.3 Architecture-Specific XAI Robustness

The convergence of multiple XAI methods (LIME, SHAP, Grad-CAM) toward lung parenchymal regions for COVID-Net, CheXNet, Xception, VGG-Based CNN, CNN-XGBoost, and CNN-YOLOv11 provides strong evidence of clinically appropriate feature learning across these architectures. For CheXNet, the mean CAM Intensity Ratio of 1.66 provides strong quantitative evidence of correct spatial focus. This corroborates Holzinger et al.'s [14] recommendation for multi-method XAI evaluation in clinical AI systems.

8.4 Impact of Dataset Balancing and Preprocessing

The VGG-Based CNN achieved the most balanced macro F1 (0.77) despite a lower AUC (0.8536), directly attributable to explicit dataset balancing. This underscores that preprocessing decisions can be as impactful as architectural choices. The shared preprocessing pipeline for CheXNet, EfficientNet-B4, and Xception ensures their AUC differences are attributable to architecture alone.

8.5 Comparison with Original Papers

Performance figures in this study are generally lower than those reported in the original papers. This gap arises from: (1) dataset differences—the RSNA dataset includes the ambiguous "No Lung Opacity / Not Normal" class not present in the original evaluation datasets; (2) frozen backbone for the ViT (vs. full fine-tuning in Shome et al. [2]); (3) binary adaptation of multi-class architectures; and (4) lower input resolution for CNN-XGBoost/YOLOv11 (64×64 vs. higher in reference papers).

8.6 Clinical Deployment Recommendations

Based on our comprehensive analysis, we provide the following deployment recommendations:

- **Best classification + XAI balance:** CheXNet (AUC=0.8804, Pointing Game=40%, CAM Intensity Ratio=1.66).
- **Best classification performance:** EfficientNet-B4 (AUC=0.8948); use LIME alongside Grad-CAM for explanation.
- **Best probability calibration:** Xception with Focal Loss (optimal threshold≈0.50, macro F1=0.81).
- **Best for resource-constrained settings:** MobileNetV2 or VGG-Based CNN, offering competitive performance with low parameter counts.
- **Best overall accuracy:** CNN-XGBoost hybrid (80.11%).
- **Threshold calibration:** Default 0.50 is suboptimal for CheXNet and EfficientNet-B4; F1-optimal thresholds significantly improve clinical sensitivity for these models.

9 Conclusion

This paper presented the most comprehensive comparative study to date of deep learning architectures for pneumonia detection from chest X-ray images, spanning nine models and ten XAI techniques on the RSNA Pneumonia Detection Challenge dataset. Our key findings are:

1. **CNN-XGBoost achieves the highest accuracy** (80.11%), demonstrating the value of hybrid CNN-classical-ML pipelines.
2. **EfficientNet-B4 achieves the highest AUC** (0.8948); its compound-scaled 380×380 resolution is the primary contributor.
3. **Xception with Focal Loss achieves the best macro F1** (0.81) and the best-calibrated probability outputs.
4. **CheXNet provides the best XAI-classification balance** (AUC=0.8804, Pointing Game=40%, CAM Intensity Ratio=1.66), making it the most suitable for clinically deployed explainable pneumonia detection.
5. **LIME is more robust than Grad-CAM** for architectures with aggressive spatial downsampling. SHAP and IG provide complementary pixel-level attributions.
6. **Multi-method XAI is essential:** No single XAI technique provides a complete picture; gradient-based and perturbation-based methods are complementary and should both be applied in clinical validation pipelines.

Future Work: (1) Full fine-tuning of the ViT backbone on medical imaging data; (2) exploring Grad-CAM++ and Score-CAM for compound-scaled architectures; (3) clinical user studies evaluating whether multi-method XAI overlays improve radiologist confidence; (4) multi-class detection distinguishing viral pneumonia, bacterial pneumonia, COVID-19, and normal; (5) prospective clinical validation under FDA SaMD guidance.

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